Time Series Databases for Digital Twin

Carlo Fabrizio · 16/03/2026 · MLG

Walloon Excellence in Technology

WHO AM I

CARLO FABRIZIO

- **BSc in Electronic and Computer Engineering**
  Università degli Studi di Trieste

- **MSc in Computer Science and Engineering** – AI focus
  Politecnico di Milano

- **PhD in Computer Science**
  Université libre de Bruxelles - MLG

«Computational Models of Memory for Streaming Data»

Digital Twins and Databases

Physical

UNLIMITED SOURCE OF DATA

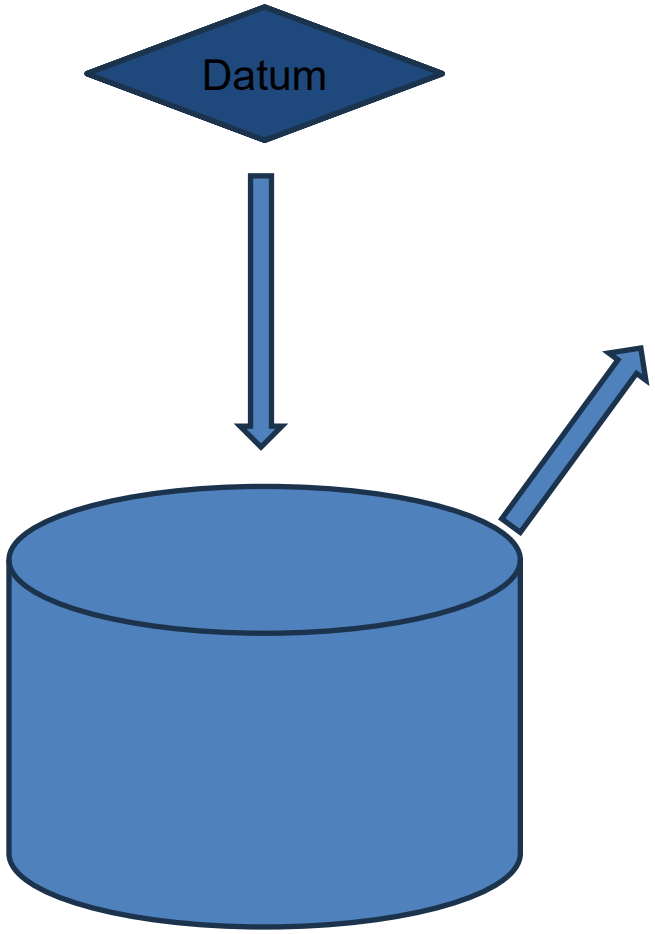
Data

Data

Digital

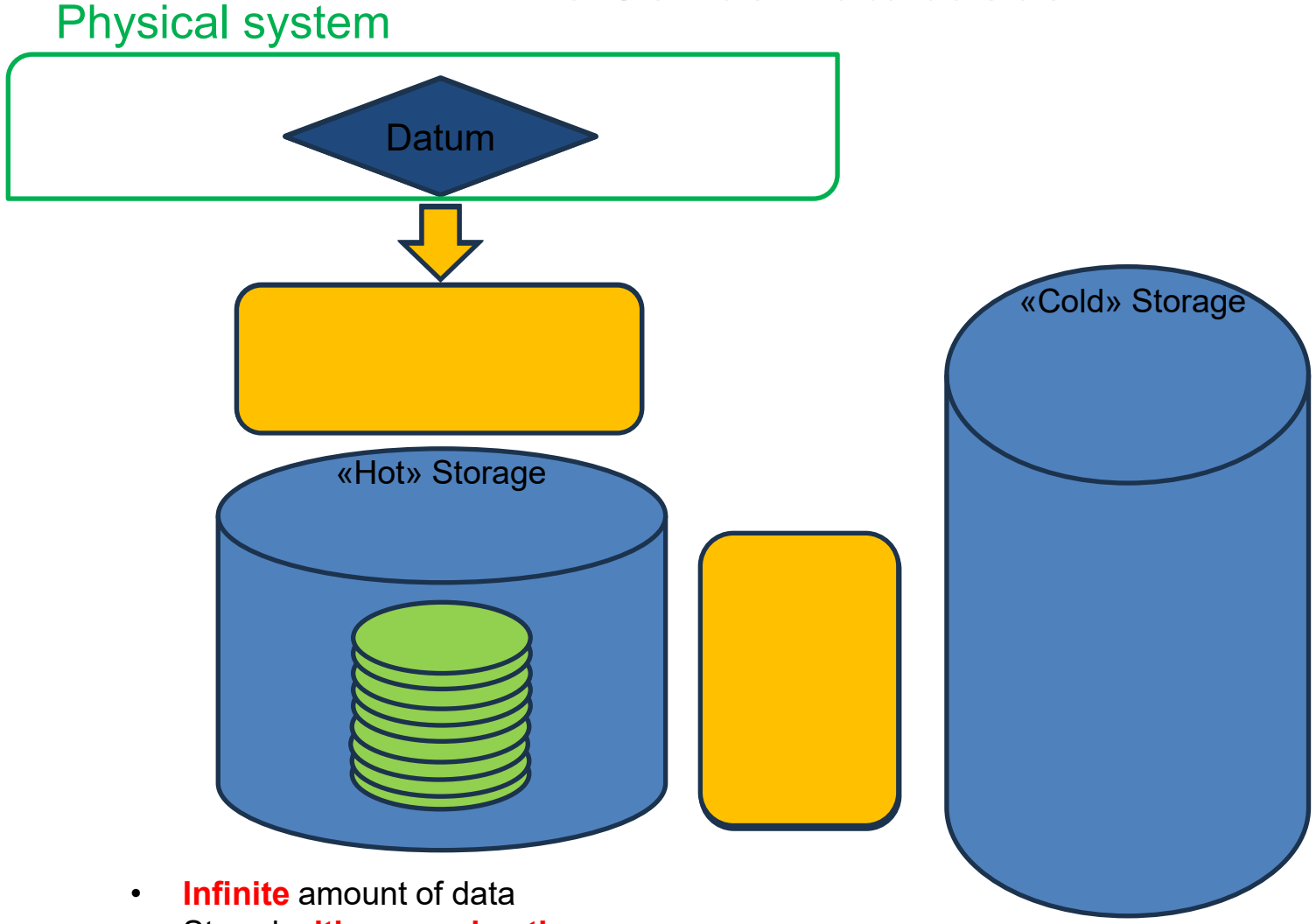
INBOUND VS OUTBOUND

Standard Databases



- **Finite** amount of data
- Stored **without approximation**

Time Series Databases



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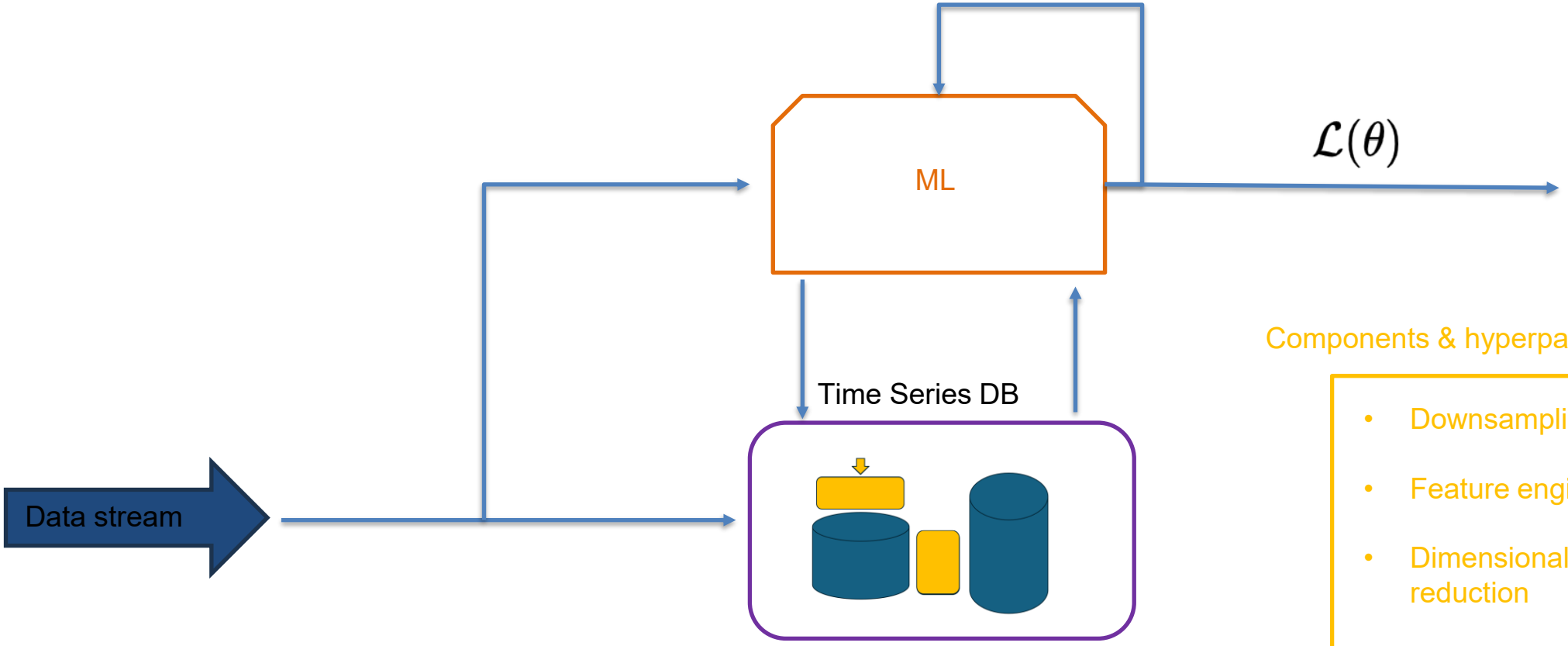
ADAPTIVE TIME SERIES DB



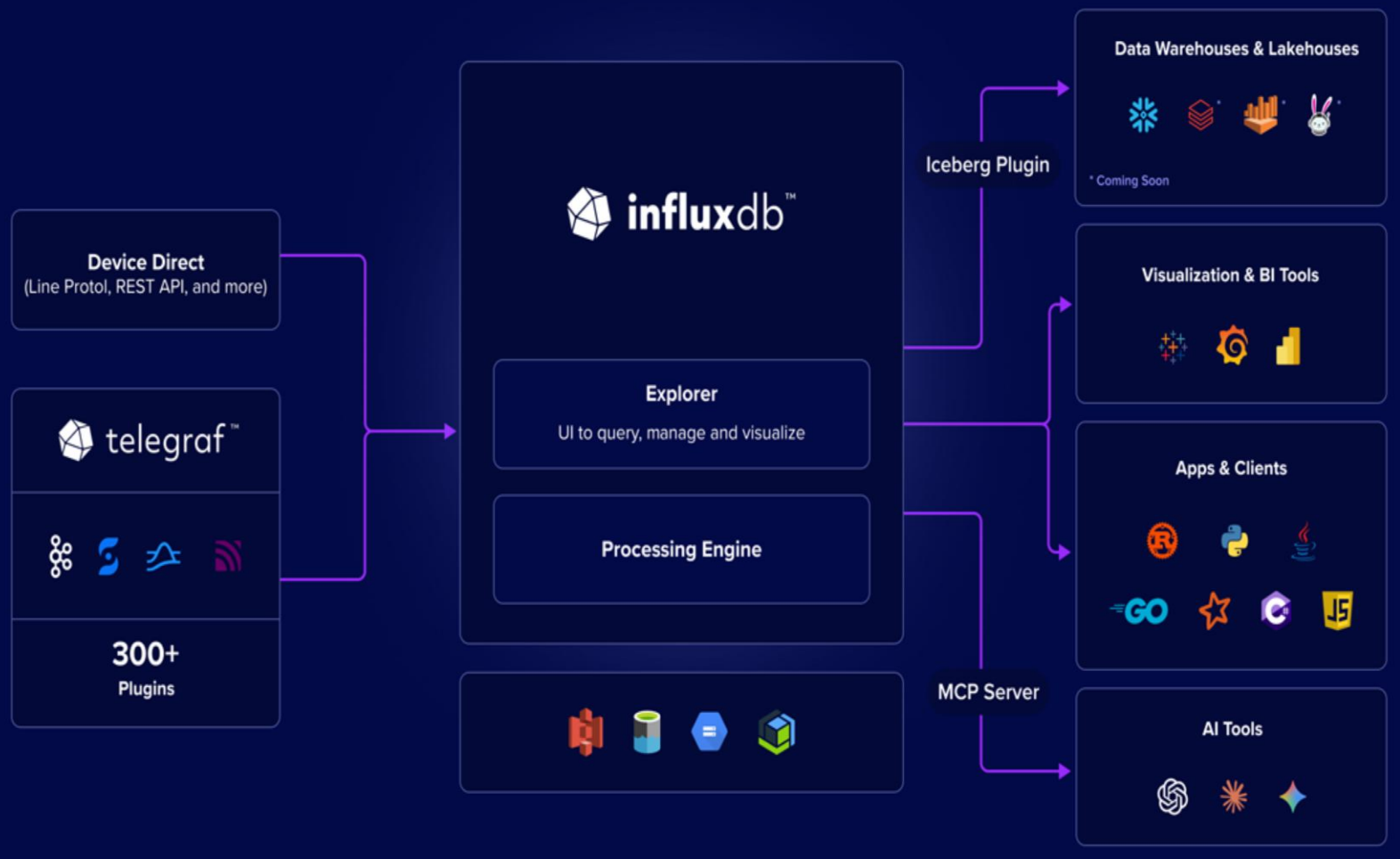
Components & hyperparameters to tune

- Downsampling
- Feature engineering
- Dimensionality reduction
- Retention policies

EXISTING TSMS

	TimeScaleDB	InfluxDB (most popular)
Retention	**Non-natively implemented**	**Easy to define**
Downsampling	**Low flexibility**	**High flexibility**
Analytics functions	**Basic**	**Advanced**: Forecasting & Anomaly Detection
Read/Write performance	⭐	⭐⭐⭐
Considerations	**Good for small use cases**	**Perfect for large-scale monitoring systems**

INFLUX DB



The Time Series Database Platform Optimized for Speed and Scale

Energy & Utilities

Gain real-time insights into turbines, refineries, meters, and equipment, and create predictive models using real-time and historical data with InfluxDB.

Learn more →

IoT Monitoring Data Platform for Digital Transformation

The enterprise IoT solution provides real-time insights and analytics into devices and processes—collecting, storing, and visualizing data from sensors.

Learn more →

InfluxDB for Network & Infrastructure Monitoring

Gain visibility into the network performance of mission-critical systems. Resolve connectivity failures, speed degradations, and the like before users notice.

Learn more →

Consumer IoT

The consumer IoT solution using InfluxDB collects, stores, and visualizes data from sensors, providing customers with analytics and insights into devices.

Learn more →

#1 Time Series Database

Source: DB Engines

ROCKETLAB · intuit · NVIDIA · Capital One · LEOLABS · Unity · SIEMENS energy

FUTURE PLAN

GOAL

Avoid catastrophic forgetting
&
handle dataset shifts

Lifelong Learning (LL)

- 6 months
  Implementation of the Adaptive Time Series DB (MSc Thesis Emanuela Centra)
  Deep study of Lifelong Learning and applications

- 12 months
  Application of Lifelong Learning to Digital Twins

Thank You